Venkata Lakshmi Parimala Pasupuleti

Lead AI Engineer · LangGraph Multi-Agent · LLM Evaluation · GenAI Platform · RAG · Agentic HITL · AWS · Production AI

linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | AgentProbe (live) |

Virginia, USA · F-1 STEM OPT → H-1B

PROFESSIONAL SUMMARY

Software Engineer and AI platform builder with 8+ years of production engineering and hands-on GenAI systems. Built **AgentProbe** -- a CI-integrated LLM evaluation platform (5-metric LLM-as-judge engine, 3-agent LangGraph runtime, multi-provider routing across Anthropic/OpenAI/Groq/Ollama, React observability dashboard -- regressions blocked before merge). Built **Leasel** -- an 8-agent LangGraph StateGraph with real conditional edges, confidence-gated HITL escalation, safety guardrails enforced in code, and 100% citation correctness on Phase 1 eval. Shipped Entrata's first production AI pipeline (PyTorch, 1.6M financial records, quality gates and compliance controls at launch). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). Targeting AI Foundations / GenAI platform roles building the evaluation, orchestration, and agent infrastructure that engineering teams depend on.

CORE COMPETENCIES

LangGraph Multi-Agent | LLM Evaluation (LLM-as-Judge) | GenAI Platform Engineering | RAG Pipelines | Multi-Provider LLM Routing | Agentic HITL Design | Foundation Model APIs | Safety Guardrails | Eval Frameworks & CI Integration | Agent Observability | Production AI Systems | AWS Infrastructure

EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Mission Planning · Compliance RAG · Safety Guardrails

- Built an agentic mission-planning system as a **LangGraph stateful graph** (parse → plan → retrieve → validate → conditional routing) -- compliant missions route to export, non-compliant ones route to rejection with grounded reasons; enforced risk-aware decision boundaries before MAVLink/QGroundControl export.
- Implemented a **RAG layer** over FAA Part 107 and operational-guidance corpus -- every compliance check functions as a hypothesis test cited to source regulation; designed backend-agnostic retriever with documented ChromaDB/embedding upgrade path.
- Engineered **safety guardrails** enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks -- blocking non-compliant missions deterministically before any agent action is taken.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

ML Research · Statistical Modeling · Anomaly Detection · Teaching (600+ students)

- Applied **Isolation Forest** to automate data integrity checks across 3 research pipelines -- surfaced anomalies manual review missed; built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Production AI · Microservices · AWS · Financial Systems at Scale

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) -- automated data processing workflows replacing manual Excel operations using a custom neural network; output validation, quality gates, and compliance controls enforced at launch.
- Built and shipped **20+ full-stack production modules** (Python, PHP, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users -- 99.5% uptime sustained across 6+ consecutive months.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines (encrypted 100% of resident phone numbers and SSNs); zero security incidents; updated 50%+ of product codebase for encryption compliance.
- Led **microservices architecture** redesign using Docker -- 50% improvement in system scalability, 30% reduction in deployment time; converted 20% of product modules into service-based architecture.
- Optimized **200+ SQL queries** (30% performance gain) through execution plan tuning, indexing, and partitioning; eliminated 80% of manual operational workload via Python/Bash automation pipelines.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · AWS S3

- Built full-stack modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts processing 1.6M records; migrated thousands of client documents to AWS S3 and redesigned document storage database architecture.

PORTFOLIO PROJECTS

AgentProbe — CI-Integrated LLM Evaluation Platform

LLM Eval | Multi-Agent | GenAI Platform

2024–Present · Python, LangGraph, LangChain, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **CI-integrated 5-metric LLM-as-judge evaluation pipeline** (factual accuracy, hallucination, retrieval, completeness, compliance) -- every commit scored automatically via GitHub Actions, regressions blocked before merge.
- Deployed a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool-use patterns, inter-agent state handoffs, and **multi-provider routing** across Anthropic, OpenAI, Groq, and Ollama -- 15-endpoint FastAPI backend running 100% locally at zero cost.
- Delivered **trace-level observability** across concurrent agent test suites -- per-run metrics, regression drift, retrieval quality, and latency -- via React dashboard backed by a ChromaDB RAG knowledge retrieval pipeline.

Leasel — 8-Agent LangGraph Platform with HITL & Safety Guardrails

Agentic HITL | Eval Pipeline | Guardrails

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite, Streamlit · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed an **8-agent LangGraph StateGraph** with real conditional edges and confidence-gated HITL escalation (Triage → Retrieval → Config → History → Structured Data → Reasoning → Validation → Human Escalation) -- correctly traces false-ineligibility to misconfigured setting and release tag in end-to-end validation.
- Enforced **safety guardrails in code**: LLM extracts and reasons; deterministic logic or human review makes every final regulated call -- every answer cited to exact source or explicitly flagged as unsupported; 0 safety boundary violations in eval.
- Built a hand-rolled TF-IDF RAG pipeline (no LangChain, no vector DB -- deliberate) achieving **100% citation correctness and 0% unsupported-claim rate** on Phase 1 eval set; Streamlit Engineer Console with 4-tab observability into ticket investigation, document Q&A, and settings.

AI on FHIR — NLP-Driven Medical Records Query System

LLM Production | NLP

2024 · Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings -- HIPAA-compliant RBAC enforced at the query layer.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Multi-Agent Orchestration, HITL, Tool Calling, Agent Tracing, Agent Observability, Safety Guardrails, Prompt Engineering

LLM Evaluation & GenAI: LLM Evaluation, LLM-as-Judge, Eval Frameworks, GenAI, Foundation Model APIs, LLM Inference, Multi-Provider LLM Routing, RAG Pipelines, ChromaDB, Vector Database, Embeddings, Semantic Search, OpenAI API, Anthropic API, Groq, Ollama

Backend & Infrastructure: Python, FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, Microservices, ETL, RabbitMQ, Linux, Libsodium

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, Random Forest, NLP, spaCy, BERT, Hugging Face, NumPy, Pandas, SQL, Feature Engineering, Statistical Modeling

Languages: Python (primary), PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash